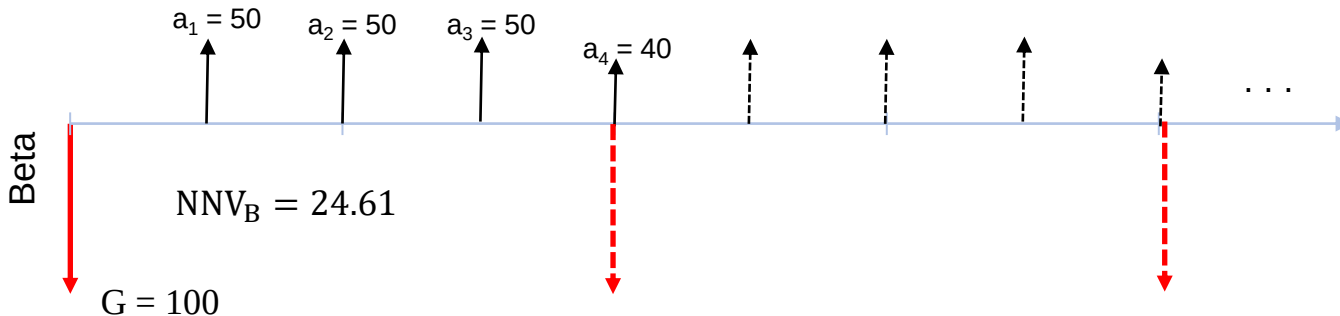
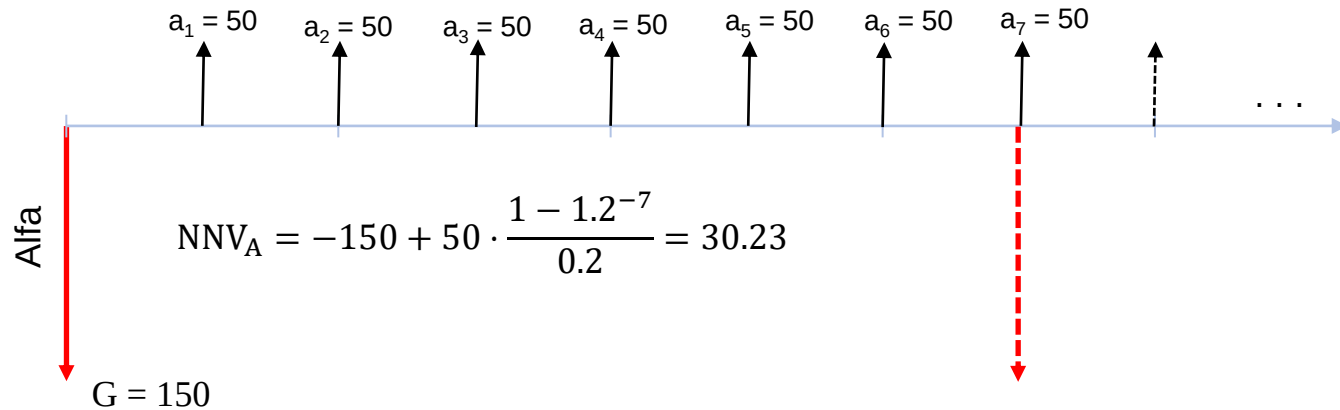


Annuitet (rak)

Investeringer med ulike N_e



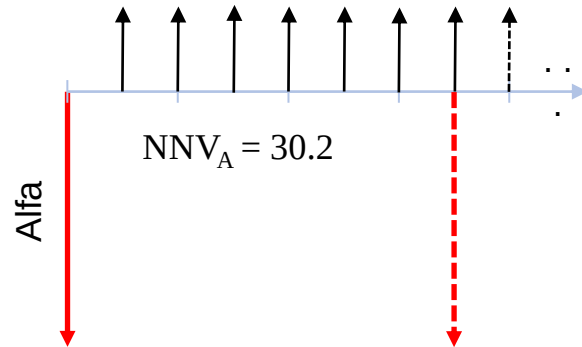
Annuitet (rak)

- Sammenlign investeringer med forskjellige levetider som kan gjentas
- Annuitet er en alternativ kontantstrøm med samme Nettonåverdi

$$NNV = k_0 + \sum_{t=1}^{N_e} \frac{k_t}{(1+r)^t} = \sum_{t=1}^{N_e} \frac{\text{annu}}{(1+r)^t}$$

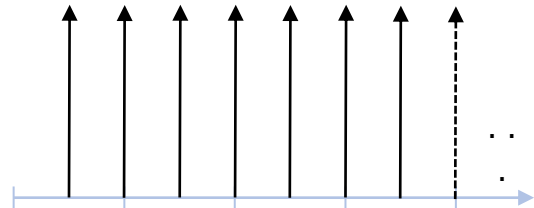
$$\text{annu} = NNV \cdot \frac{r}{1 - (1+r)^{-N_e}}$$

- Vid samme r - jo høyere annuitet jo bedre
- Invester hvis annuiteten er positiv



$\Leftrightarrow$

$$\text{annu}_A = \frac{0.2}{1 - 1.2^{-7}} \cdot 30.23 = \mathbf{8.39}$$



Nettonåverdi – Uendelig serie

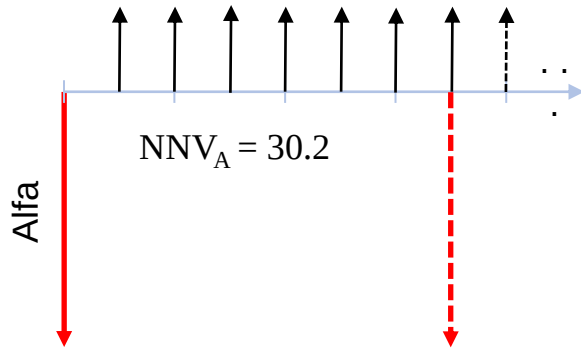
- Konstant kontantstrøm, eks. rak annuitet :

$$NNV = \lim_{N \rightarrow \infty} k \sum_{t=1}^N \frac{1}{(1+r)^t} = \lim_{N \rightarrow \infty} k \cdot \frac{1 - (1+r)^{-N}}{r} = \frac{k}{r}$$

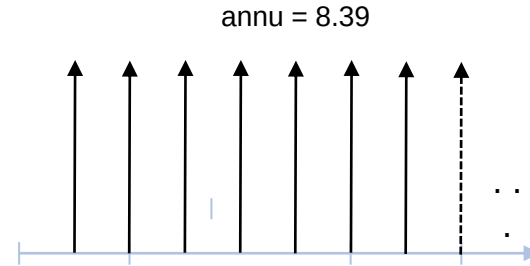
- Eksponentielt økende, proporsjonal (%) tilvekst, eks. inflasjon, gotteri og gull

$$NNV = \lim_{N \rightarrow \infty} k \sum_{t=1}^N \frac{(1+g)^{t-1}}{(1+r)^t} = \frac{k}{r-g}$$

Nettonåverdi – uendelig serie



$\Leftrightarrow$



$$NNV_{tot} =$$

$$\sum_{Gen=0}^{\infty} \frac{k_0}{(1+r)^{Gen \cdot N_e}} + \sum_{t=1}^{N_e} \frac{k_{Gen \cdot N_e + t}}{(1+r)^{Gen \cdot N_e + t}}$$

$$NNV_{tot} =$$

$$\sum_{t=1}^{\infty} \frac{annu}{(1+r)^t} = annu \frac{1 - (1+r)^{-\infty}}{r} =$$

$$\frac{8.39}{0.2} = 41.9$$

Avledning av $NNV = k/(r-g)$

$$NNV = \lim_{N \rightarrow \infty} k \sum_{t=1}^N \frac{(1+g)^{t-1}}{(1+r)^t} = \lim_{N \rightarrow \infty} \frac{k}{1+g} \sum_{t=1}^N \frac{1}{((1+r)/(1+g))^t}$$

$$= \lim_{N \rightarrow \infty} \frac{k}{1+g} \sum_{t=1}^N \frac{1}{((1+g-g+r)/(1+g))^t}$$

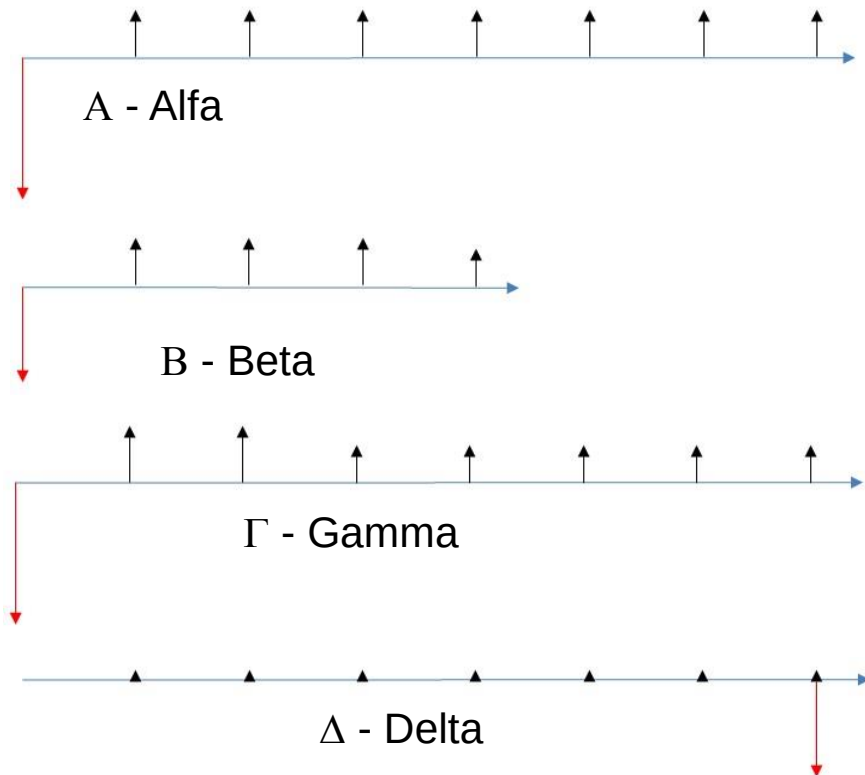
$$= \lim_{N \rightarrow \infty} \frac{k}{1+g} \sum_{t=1}^N \frac{1}{(1+(r-g)/(1+g))^t}$$

$$= \lim_{N \rightarrow \infty} \frac{k}{1+g} \frac{1 - (1+(r-g)/(1+g))^{-\infty}}{(r-g)/(1+g)}$$

$(r-g)/(1+g)$ er
den fiktive
renten r_f

Eksempel

	A	B	Γ	Δ
G	150	100	150	1
N_e	7	4	7	7
a_1	50	50	60	5
a_2	50	50	60	5
a_3	50	50	40	5
a_4	50	40	40	5
a_5	50		40	5
a_6	50		40	5
a_7	50		40	5
S	0	0	0	-100
r	20%	20%	20%	20%



Annuitet (rak)

	A	B	Γ	Δ
G	150	100	150	1
n_e	7	4	7	7
a_1	50	50	60	5
a_2	50	50	60	5
a_3	50	50	40	5
a_4	50	40	40	5
a_5	50		40	5
a_6	50		40	5
a_7	50		40	5
S	0	0	0	-100
r	20%	20%	20%	20%

$$annu = NNV \cdot \frac{r}{1 - (1 + r)^{-N_e}}$$

$$annu_A = 30.23 \cdot \frac{0.2}{1 - 1.2^{-7}} = 8.39 \quad \Rightarrow \frac{8.39}{0.2} = 41.93$$

$$annu_B = 24.61 \cdot \frac{0.2}{1 - 1.2^{-4}} = 9.51 \quad \Rightarrow \frac{9.51}{0.2} = 47.54$$

$$annu_{\Gamma} = 24.74 \cdot \frac{0.2}{1 - 1.2^{-7}} = 6.86 \quad \Rightarrow \frac{6.86}{0.2} = 34.32$$

$$annu_{\Delta} = -10.89 \cdot \frac{0.2}{1 - 1.2^{-7}} = -3.02 \quad \Rightarrow \frac{-3.02}{0.2} = -15.10$$